

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered **distributed executor** routing nodes to **Docker** or **AWS Lambda**; content-addressed **S3** artifacts make re-runs **idempotent**.
- Designed crash recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step**.
- Provisioned and operated **multi-cloud** clusters (**AWS, GCP, CoreWeave**) with **InfiniBand/RoCE** networking for **high availability**.
- Wrote systems code in **Go** and **C++** for **scheduling, resource allocation, and fault-tolerant placement**.
- Built **SSE** observability over live **metrics and logs** with **automated incident detection**; cut redundant API calls **90%**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Served in **on-call rotation**, resolving **distributed system incidents** end-to-end; built **CI/CD rollback** and **failure alerting**.
- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** via async serving on **GCP Cloud Run** with **Kubernetes**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ C++ microservices** in SQL-backed **distributed systems** for **1M+ users** at **99.9% uptime** on **AWS**.
- Standardized failure handling across **25+ services**, cutting escalations **60%** via consistent **retry and timeout** semantics.
- Profiled **compute/memory bottlenecks**, optimizing **batching** and scheduling in **Python/C++** services; **throughput 2.4x**.
- Cut nightly aggregation blocking **settlement** from **8 min to 90 sec** via **indexed views** and **batch partitioning**.
- Drove **root cause analysis** and **post-incident reviews** for failures spanning multiple services and **SQS** queues.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **leader election**, **linearizable reads**, and **log compaction**; **VPC control plane** simulator.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system diagnosing **production incidents** from alerts; synthesizes **root cause reports**, cutting **MTTR**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Observability platform** with **regression detection** and **latency tracking**; **FastAPI, React/TypeScript, PostgreSQL**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Go, Python, Java, SQL, TypeScript, HTML/CSS

Distributed Systems: Raft Consensus, Linearizable Reads, Log Compaction, Idempotent Retries, Checkpointing, Messaging, DS&A

Cloud & Infra: AWS (EC2, S3, Lambda, RDS, SQS, EKS), GCP, Azure, CoreWeave, Docker, Kubernetes, Terraform

Observability & Data: Prometheus, Grafana, Redis, PostgreSQL, On-call, Root Cause Analysis, Post-incident Reviews

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer